Week 2 Delivery to a target

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A 40 cm delivery

Predict together Measure C = ____ cm around our wheel. Predict 180 degrees: ____ cm; 360 degrees: ____ cm. Explain using half and one full turn.

40 cm target: (40 / C) × 360 = _____ wheel degrees. Round at the end.

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Predict cm = (degrees / 360) × C. Use our measured C and the full supplied project.

Run	Wheel degrees	Predicted cm	Measured cm
1			
2			
3			

Build A cargo tray that keeps its load

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about reaching a target? What did you observe in your distance tests?

Next idea _____ Saved as _____

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Week 2 build card

A cargo tray that keeps its load

Team ____ Builders A ____ B ____ C ____

Build a removable cart/tray with an edge that keeps the cube aboard. **It must not hinder the robot's movement.**

Materials: Beams and pins or folded pre-cut card, tape, one lightweight cube, and the mounting/clearance template.

Try it

1. Choose where the cube will sit and build a flat support.

2. Add an edge or rail without making the tray touch the floor or wheels.

3. Tilt the tray gently by hand and compare one change to its edge.

Our jobs

Builder connects pieces. Designer chooses/finds parts and checks the plan. Tester performs the hand test and reports one result. Everyone builds. Rotate when the mentor calls time.

Test and improve

Cube stays aboard during a gentle tilt. The cart fits securely, clears the wheels and floor, and cannot snag cables or block controls. A mentor checks a short run; if it rubs, drags or shifts, fix the build.

We noticed _____. We changed _____.

A mentor supplies an example tray for the first robot turn. Keep payload and tray unchanged during each distance comparison; test the team tray during the retest.

If time allows: after the core trials, attach the checked cart for the book-parking challenge. Help the programmers measure from its frontmost part and leave a gap before the book. The mentor chooses and supervises the tabletop route.

Fit attachments only to a stopped robot after the mentor checks clearance. Leave wheels, buttons, cables and sensors clear.

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Wheel turns and distance

Team ____ Kit ____ Our measured circumference C = ____ cm

Around the wheel

Diameter d goes across the center. Radius r is half the diameter. Circumference C goes all the way around the tire.

C = pi × d = 2 × pi × r. Pi is about 3.14.

With the robot off, wrap a string or paper strip once around the tire. Straighten it and measure the length. Ask the mentor for help.

Part of a turn

90 degrees = quarter turn. 180 degrees = half turn. 360 degrees = one full turn. 720 degrees = two full turns.

One full wheel turn predicts one circumference of forward travel when the wheel rolls straight without slipping.

Code to distance

Predicted distance = (wheel degrees / 360) × C

Example only, C = 17.6 cm: 180 degrees predicts 8.8 cm; 360 predicts 17.6 cm; 720 predicts 35.2 cm.

For OUR kit, 180 degrees predicts ____ cm and 360 predicts ____ cm.

Distance to code

Wheel degrees = (target distance / C) × 360

Example only: for 40 cm with C = 17.6 cm, (40 / 17.6) × 360 is about 818 degrees. Round only at the end.

For OUR kit, (40 / ____) × 360 = ____ wheel degrees.

Use centimeters for both target and C. One meter = 100 cm. The 1 meter question is a paper calculation today.

Our test

Change wheel_degrees in the supplied program. Keep speed, floor, tray, cube and start direction the same. Measure movement of the same chassis mark. Test 180, then 360, then your calculated 40 cm value. Stay within 0–1080 degrees and ask the mentor before running.

The motor counts turns. It does not measure distance across the floor. Measure the real travel and explain any difference.

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Session attendance and delivery record

Session number _____ Date _____ Actual start _____ Actual end _____

Adult lead _____ Mentors _____

Student ID	Minutes present	Language used	Driver turn confirmed	Notes
S01				
S02				
S03				
S04				
S05				
S06				
S07				
S08				
S09				
S10				
S11				
S12				

Use language labels Python, Word Blocks, or Unplugged. A short observation can identify one concept explained or an access adjustment. Keep medical and family information out of this sheet.

Unique students today _____ Sum of participant minutes _____ / 60 = contact hours _____

Mentor preparation hours _____ Mentor delivery hours _____ Adult hours _____

What worked and what should change next week

_____

_____

_____

Equipment or file issue to resolve before next session

_____

_____